Eigenvalues and eigenvectors: the directions a matrix only stretches

In a table with many columns, which directions do the data vary along most, and how do we look at embeddings with hundreds of numbers?

Mathematics for AI and Machine Learning | Session 5 of 18 | Linear algebra

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Today in one view: what you will leave with

By the end of today you can



Say what an eigenvector and an eigenvalue are, with a picture of a matrix acting on a circle

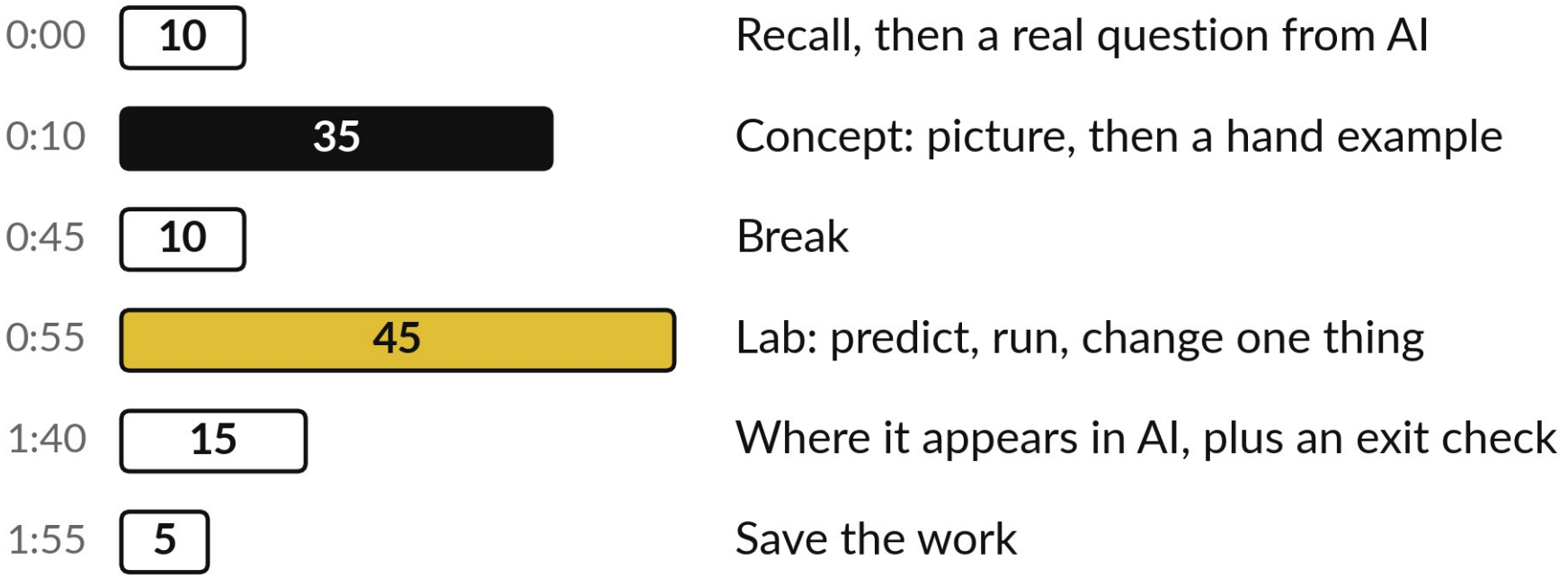


Find the eigenvalues and eigenvectors of a 2 by 2 matrix by hand and check them with trace and determinant



Build a covariance matrix, find its top direction, and use power iteration to find it by repeated multiplying

Your two hours: stretch directions, then data



Ideas first, with pictures. Then the lab in Python or in a spreadsheet. Then where it shows up in AI.

Three questions from last time

1. What does a determinant of 0 tell you about a matrix?

2. Write the normal equation for fitting a line.

3. What is the rank of the matrix with rows (1, 2) and (2, 4)?



Answer aloud first. Then we check together.

Answers to the three questions from Session 4

Question	Short answer
What does a determinant of 0 tell you about a matrix?	It squashes the plane onto a line or a point. It has no inverse and its rank is below full.
Write the normal equation for fitting a line.	$X^TX\,w = X^Ty$. It gives the weights w with the smallest squared error.
What is the rank of the matrix with rows (1, 2) and (2, 4)?	1. The second column is twice the first, so the columns give only one direction.

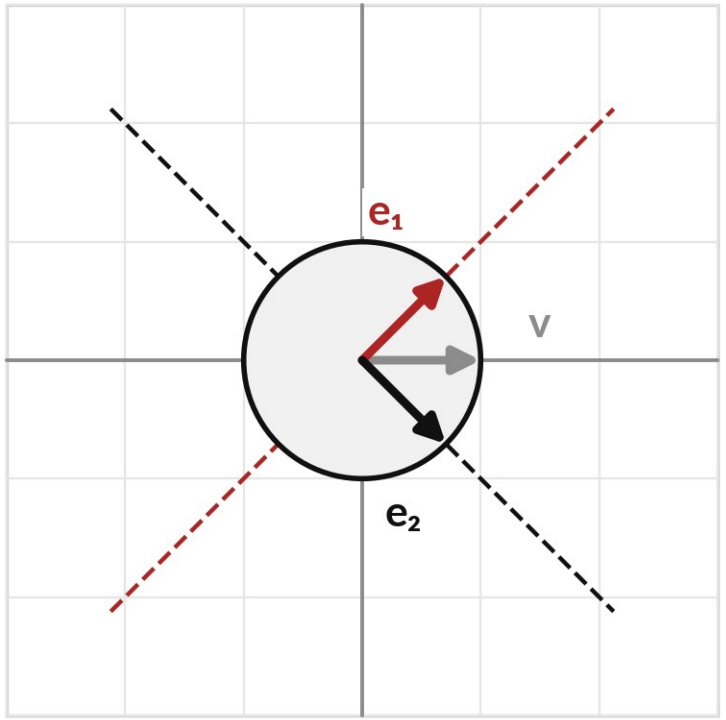
If one surprised you, look back at Session 4 before the lab. Rank and determinant return today.

Words we use today

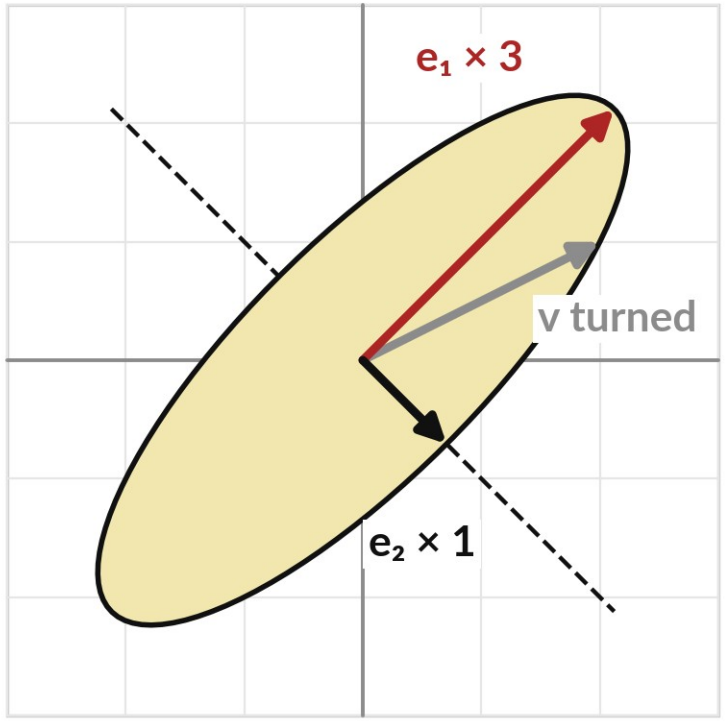
	Word	What it means
	**Eigenvector**	A direction that a matrix only stretches, flips or squashes. It stays on its own line.
	**Eigenvalue**	The factor by which an eigenvector is stretched. Written with the Greek letter lambda (λ).
	**Trace**	The sum of the diagonal entries of a square matrix.
	**Covariance matrix**	A table of how every pair of features varies together. The diagonal holds each feature's variance, its average squared distance from its mean.
	**Power iteration**	Multiplying by a matrix again and again to find its top eigenvector.

Most arrows turn when a matrix acts on them, but a few directions only stretch

Before: the unit circle



After the matrix: an ellipse



The matrix with rows (2, 1) and (1, 2) turns a circle into an ellipse. The grey arrow v turns.

e_1 points along (1, 1) and becomes 3 times longer.

e_2 points along (1, −1) and keeps its length.

Such directions are eigenvectors.

An eigenvector v and its eigenvalue λ satisfy Av = λv

Definition

$$A\mathbf{v} = \lambda\mathbf{v}$$

Our first pair

$$A\,(1, 1) = (3, 3) = 3\,(1, 1)$$

Symbol	What it means	In our data
A	A square matrix. Here the rows are (2, 1) and (1, 2).	2 by 2
v	An eigenvector: a nonzero vector whose direction A keeps.	(1, 1) or (1, −1)
λ	The eigenvalue (Greek letter lambda): the factor by which A stretches v.	3 for (1, 1), and 1 for (1, −1)
Av	Matrix times vector. For (1, 1): (2 + 1, 1 + 2).	(3, 3)

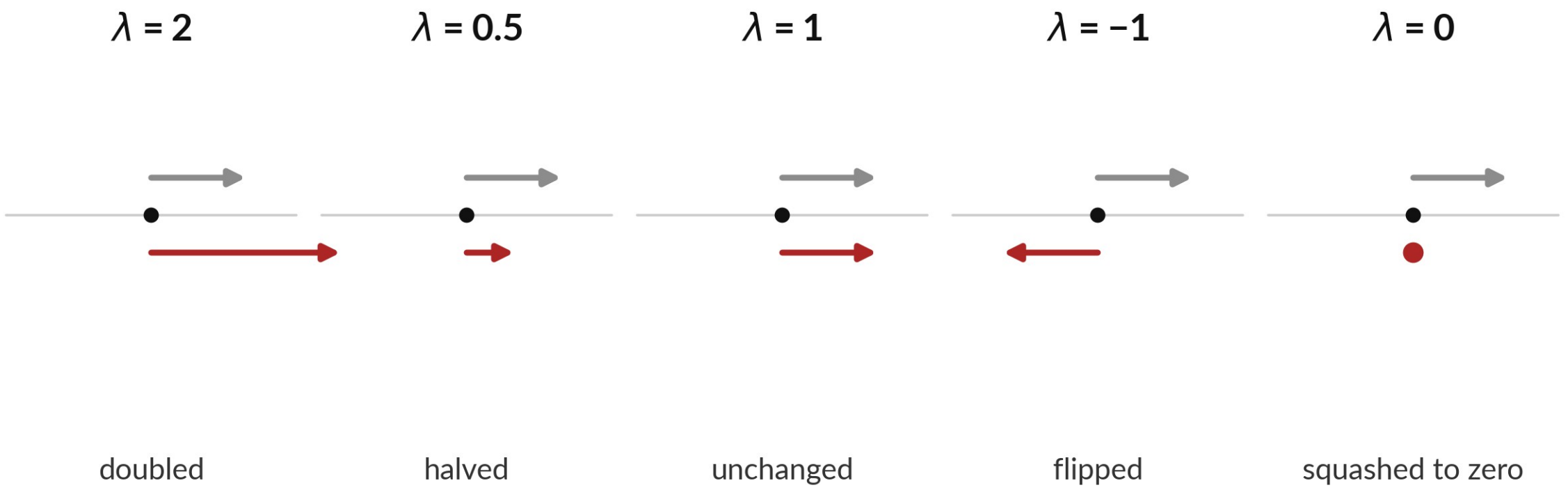
A negative λ flips v. A λ of 0 squashes it to the origin. Any multiple of v is an eigenvector too.

Testing four arrows: only some keep their direction

Arrow v	Av	Same line as v?	Stretch factor
(1, 0)	(2, 1)	No, it turned	none
(1, 1)	(3, 3)	Yes	3
(1, −1)	(1, −1)	Yes	1
(1, 2)	(4, 5)	No, it turned	none

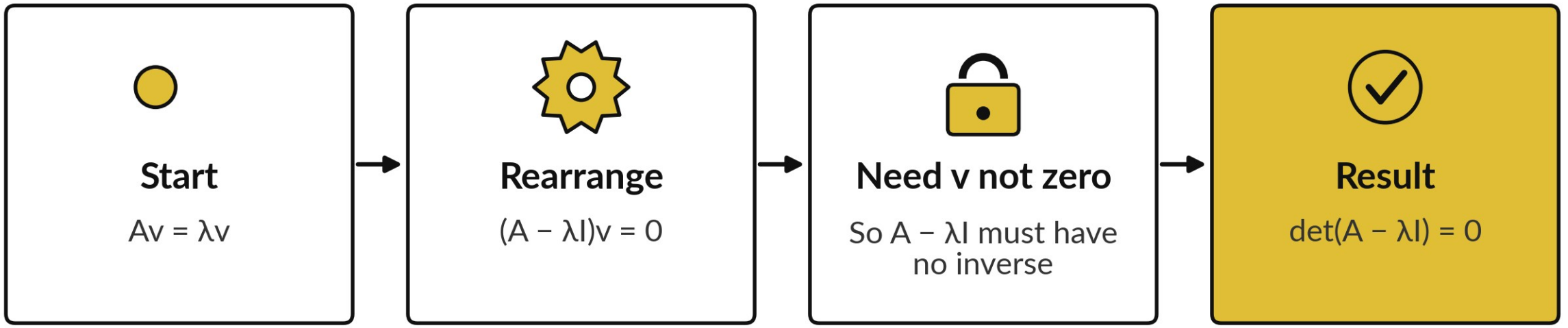
A has rows (2, 1) and (1, 2). Compute Av with a dot product for each row: for (1, 2) it is 2 + 2 = 4 and 1 + 4 = 5.

The eigenvalue says whether the arrow is stretched, shrunk, flipped or squashed



Grey arrow: an eigenvector. Red arrow: what A turns it into. λ (lambda) is the multiplier.

To find eigenvalues, ask when A − λI has no inverse



I is the identity matrix, so λI has λ on the diagonal. No inverse means determinant 0, the singular case from Session 4.

Writing det(A − λI) = 0 out for our matrix

A has rows (2, 1) and (1, 2). We subtract λ from each diagonal entry and take the determinant.

A − λI	Rows $(2 - \lambda, 1)$ and $(1, 2 - \lambda)$.
Determinant	$(2 - \lambda)(2 - \lambda) - 1 \times 1 = \lambda^2 - 4\lambda + 3$.
Set it to 0	$\lambda^2 - 4\lambda + 3 = (\lambda - 3)(\lambda - 1) = 0$, so $\lambda = 3$ or $\lambda = 1$.
Any 2 by 2	The same steps give $\lambda^2 - T\lambda + D = 0$, with T the sum of the diagonal and D the determinant.

For any 2 by 2 matrix the characteristic equation is a quadratic

Characteristic equation

$$\lambda^2 - T\lambda + D = 0$$

Eigenvalues

$$\lambda = \frac{T \pm \sqrt{T^2 - 4D}}{2}$$

Symbol	What it means	In our data
T	The trace: the sum of the diagonal entries, $a_{11} + a_{22}$.	2 + 2 = 4
D	The determinant: $a_{11}a_{22} - a_{12}a_{21}$ (a_{ij} is row i, column j of A).	2 × 2 − 1 × 1 = 3
±	Plus gives one eigenvalue, minus gives the other.	(4 + 2) / 2 and (4 − 2) / 2
λ	The two eigenvalues.	3 and 1

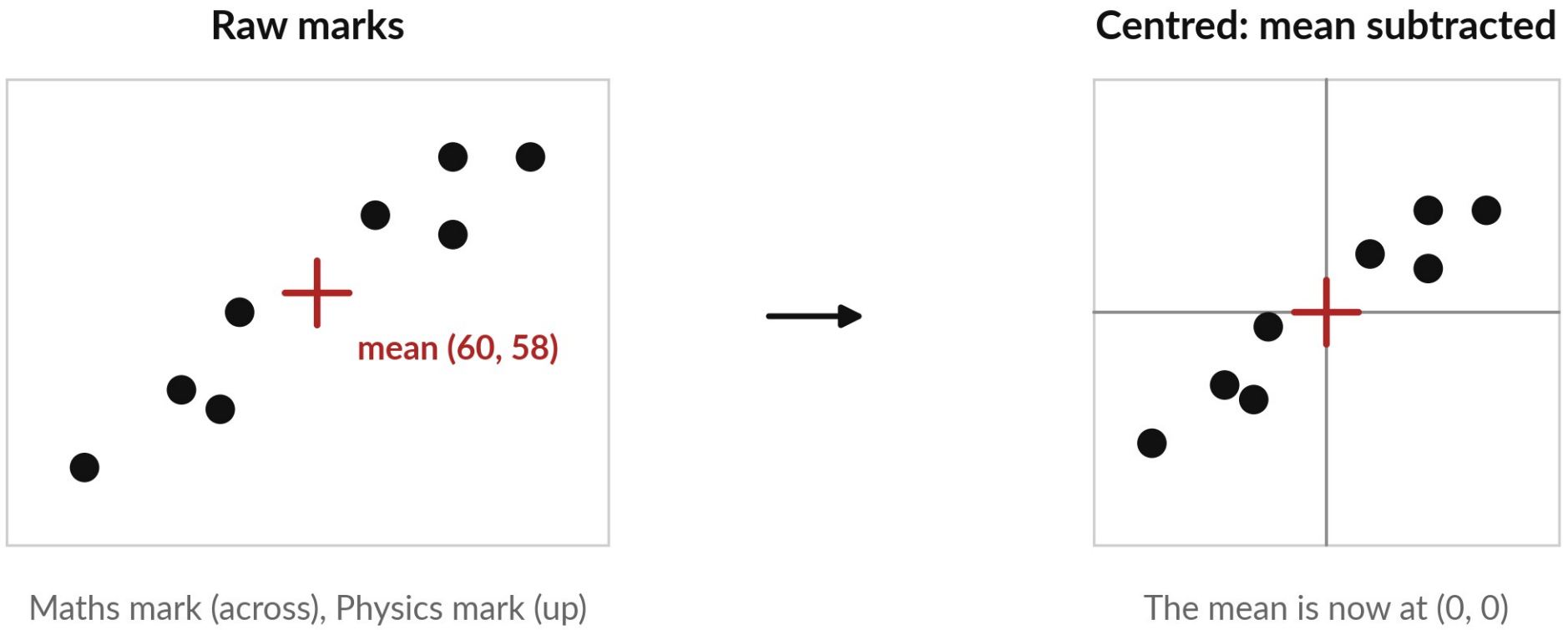
Check: the eigenvalues add up to the trace (3 + 1 = 4) and multiply to the determinant (3 × 1 = 3).

Worked example: eigenvalues and eigenvectors of our matrix

A has rows (2, 1) and (1, 2). We find λ first, then an eigenvector (s, t) for each λ.

Eigenvalues	$\lambda^2 - 4\lambda + 3 = 0$. So $\lambda = (4 + 2) / 2 = 3$ or $\lambda = (4 - 2) / 2 = 1$.
For λ = 3	$A - 3I$ has rows $(-1, 1)$ and $(1, -1)$. The equation $-s + t = 0$ gives $s = t$, so $e_1 = (1, 1)$.
For λ = 1	$A - I$ has rows $(1, 1)$ and $(1, 1)$. The equation $s + t = 0$ gives $e_2 = (1, -1)$.
Check	$A(1, 1) = (3, 3)$ and $A(1, -1) = (1, -1)$. Both match.

Centring: subtract each feature's mean so the cloud sits on the origin



Eight students (illustrative). Mean maths 60, mean physics 58. Each mark minus its mean.

The covariance matrix collects how every pair of features varies together

Covariance of x and y

$$\operatorname{cov}(x, y) = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})$$

Covariance matrix

$$C = \frac{1}{n-1} X_c^T X_c$$

Symbol	What it means	In our data
n	The number of rows, here students.	8 students
x_i, y_i, x-bar, y-bar	The two marks of student i, and the two means. Σ adds up over i = 1 to n.	means 60 and 58
X_c	The centred data: a row per student, a column per mark.	8 by 2
C	Variances on the diagonal, covariance off it. Symmetric: entry (i, j) equals entry (j, i).	2 by 2

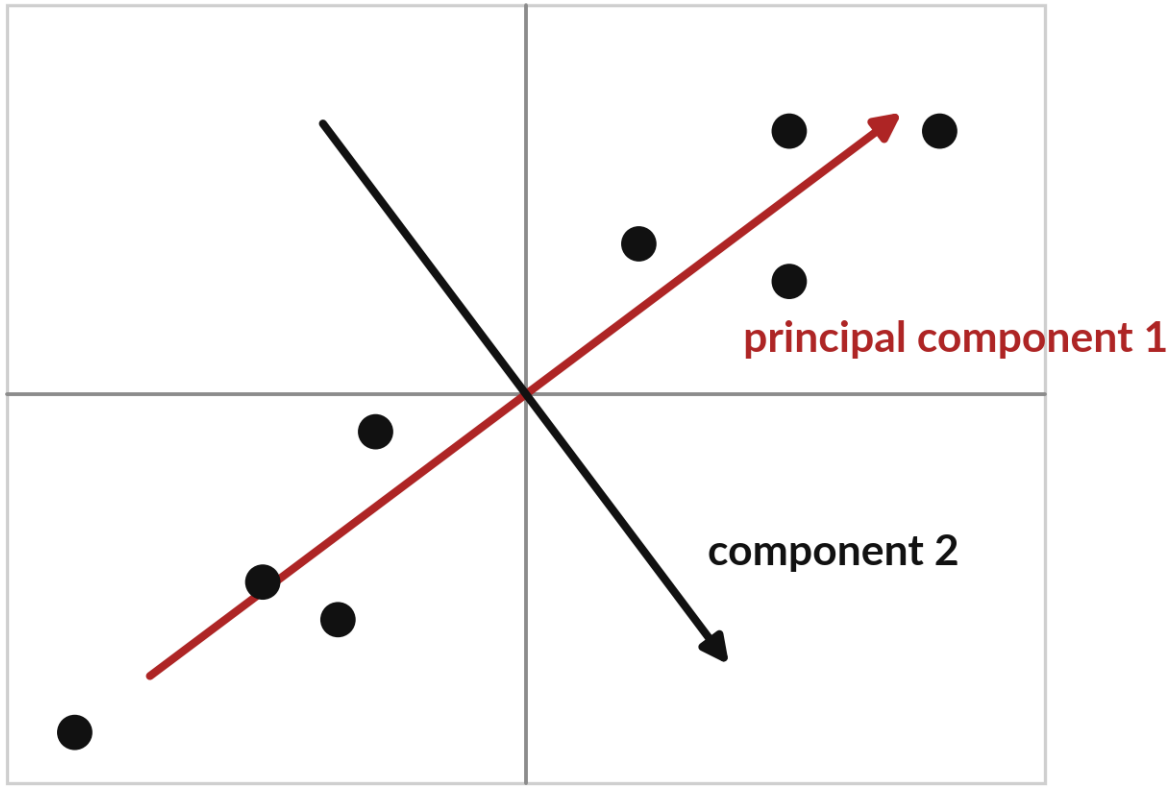
We divide by n − 1, as np.cov does. A feature replaced by 100 minus itself flips its covariance sign.

Worked by hand: the covariance matrix of the eight marks

Illustrative marks. Centred = mark minus its mean.

Maths marks	53, 55, 63, 67, 48, 67, 56, 71. Mean 60. Centred: −7, −5, 3, 7, −12, 7, −4, 11.
Physics marks	53, 52, 62, 65, 49, 61, 57, 65. Mean 58. Centred: −5, −6, 4, 7, −9, 3, −1, 7.
Sums	Maths squared 462. Physics squared 266. Maths times physics 336.
Divide	Divide by n − 1 = 7. Variances 66 and 38. Covariance 48. So C has rows (66, 48) and (48, 38).

The eigenvectors of the covariance matrix are the directions of the cloud



Centred maths mark (across) and centred physics mark (up)

Each dot is a student, centred. The red arrow is the first eigenvector of C (direction 1), the black arrow the second.

e_1 = (0.8, 0.6): principal component 1, variance 102.

e_2 = (0.6, −0.8): variance only 2.

For a unit-length eigenvector the eigenvalue is the variance along it.

Worked by hand: eigenvalues and top eigenvector of C

C has rows (66, 48) and (48, 38). We use the 2 by 2 formulas from earlier.

Trace and determinant	T = 66 + 38 = 104. D = 66 × 38 − 48 × 48 = 204.
Eigenvalues	The square root of $104^2 - 4 \times 204$ is 100. So λ = (104 + 100) / 2 = 102 or (104 − 100) / 2 = 2.
Top eigenvector	C − 102I has rows (−36, 48) and (48, −64). So −36s + 48t = 0, which gives t = 0.75s. Take (4, 3). Length 5, so e_1 = (0.8, 0.6).
Share of variance	102 / 104 = 98.1 percent of all the variance lies along direction 1.

Power iteration finds the top eigenvector by repeated multiplying

Update

$$\mathbf{g}_{k+1} = \frac{A\,\mathbf{g}_k}{\max|A\,\mathbf{g}_k|}$$

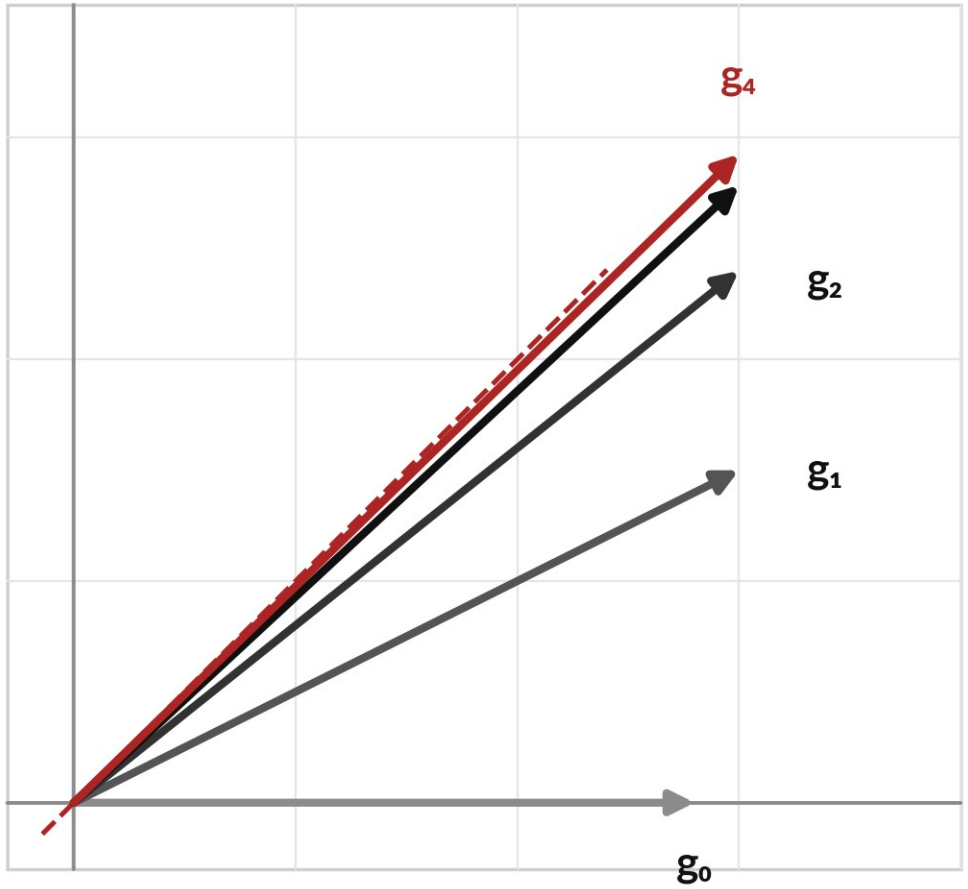
Eigenvalue estimate

$$\lambda \approx \max|A\,\mathbf{g}_k|$$

Symbol	What it means	In our data
$\mathbf{g}_k$	The current guess for the eigenvector, after k steps.	g_0 = (1, 0) to start
$A\mathbf{g}_k$	Multiply the guess by the matrix.	one matrix times vector product
max |.|	The entry with the largest size. Dividing by it keeps the numbers small.	g then has largest entry 1
λ	The top eigenvalue, the one largest in size.	estimate tends to 3

It works when one eigenvalue is largest in size and the start is not exactly along another eigenvector.

Each multiplication swings the arrow closer to the top eigen-direction



Matrix with rows (2, 1) and (1, 2). Start at g_0 = (1, 0). Multiply, then divide by the largest entry.

g_1 = (1, 0.50), g_2 = (1, 0.80), g_4 is almost (1, 1).

The estimates of λ are 2.00, 2.50, 2.80, 2.93, 2.98. They head to 3.

Power iteration by hand on the covariance matrix C

Step k	Guess g_k	C g_k	Largest entry
0	(0.000, 1.000)	(48.0, 38.0)	48.00
1	(1.000, 0.792)	(104.0, 78.1)	104.00
2	(1.000, 0.751)	(102.0, 76.5)	102.04
3	(1.000, 0.750)	(102.0, 76.5)	102.00
4	(1.000, 0.750)	(102.0, 76.5)	102.00
5	(1.000, 0.750)	(102.0, 76.5)	102.00

Start g_0 = (0, 1). The next guess is C g_k divided by the largest entry. After 6 multiplications (row k = 5) the estimate is 102.0, close to 102, and the guess is near (1, 0.75), which is (0.8, 0.6) at length 1.

Common mistake: four things to remember about eigenvectors

	Mistake	What is true	What to do
✗	**Treating an eigenvector as one fixed arrow**	Any multiple, and the opposite arrow, is also an eigenvector.	Quote a unit-length version. Expect sign flips between tools.
✗	**Looking for eigenvalues of a non-square matrix**	Only square matrices have them.	Use the SVD for other shapes (Session 6).
✗	**Expecting real eigenvalues always**	A quarter-turn rotation has no real eigenvalues, because every arrow turns.	Covariance matrices are symmetric and always give real ones.
✗	**Mixing up the two outputs**	For a covariance matrix, the eigenvalue is the spread along its eigenvector.	Quote both together.

Read the recipe before you write it: covariance, then eigenvectors

```
Xc = X - X.mean(axis=0)
C = Xc.T @ Xc / (len(X) - 1)
values, vectors = np.linalg.eigh(C)
g = np.array([0.0, 1.0])
for k in range(6):
    Cg = C @ g
    est = np.max(np.abs(Cg))
    g = Cg / est
```

In plain words

Xc = ... centres each column by subtracting its mean.

C = ... is the covariance matrix, as in the formula.

eigh is for symmetric matrices. It lists eigenvalues from smallest to largest, with eigenvectors in the columns.

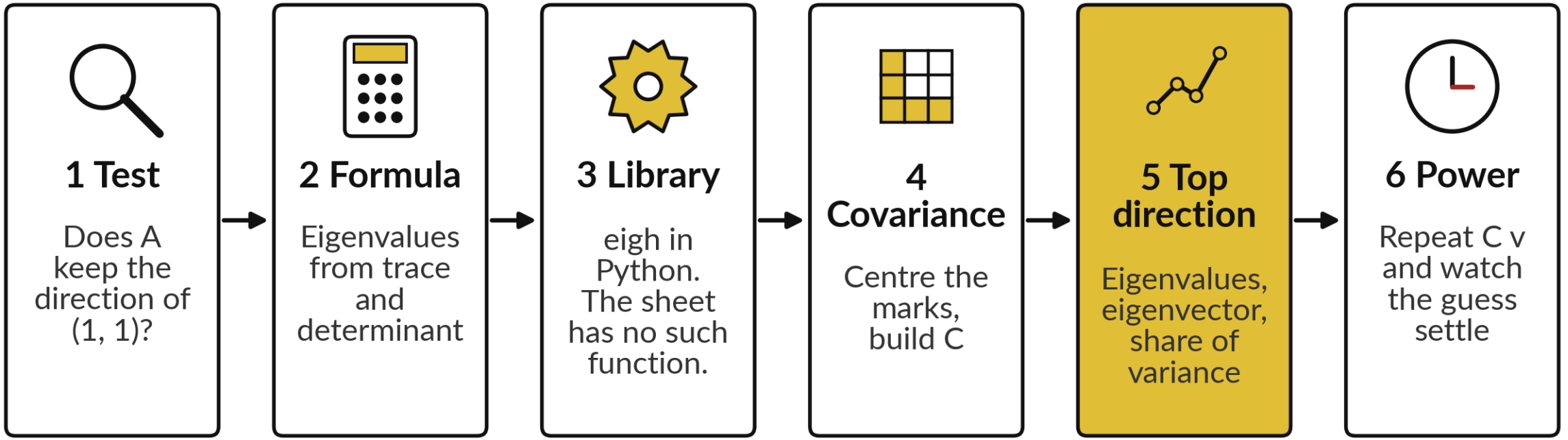
The loop from g = ... is power iteration. est holds the eigenvalue estimate.



Part 2: the lab

Close the slides. Open the notebook or the workbook.

The lab has six steps, in the notebook and in the workbook



Python uses numpy. The spreadsheet uses MMULT, VAR.S, COVARIANCE.S, SQRT and rows of cells. Gold cells are yours.

Check your numbers against these before you move on

What you compute	You should see
A times (1, 1) for the small matrix	(3, 3)
Eigenvalues of the small matrix	3 and 1
Covariance matrix C of the eight marks	rows (66, 48) and (48, 38)
Eigenvalues of C	102 and 2
Top eigenvector of C, unit length	(0.8, 0.6)
Share of variance along direction 1	0.981
Power iteration after 6 multiplications (row k = 5): largest entry	102.0
Covariance if physics is replaced by 100 minus physics	−48: the sign flips, the size stays

If a number differs, stop and find the step that went wrong. The sign of an eigenvector may differ between tools.

Four errors you will meet, what they say and what to do

	Error or symptom	What it says	What to do
	**The eigenvector has a minus sign**	The opposite arrow is also an eigenvector.	Compare absolute values, or flip the sign.
	**Eigenvalues come out in the wrong order**	eigh lists smallest first.	Reverse the list or sort it.
	**Covariance far too large**	The data were not centred first.	Subtract the column means.
	**#NUM! in the spreadsheet**	The square root of a negative number.	Check $T^2 - 4D$. For covariance it is never negative.

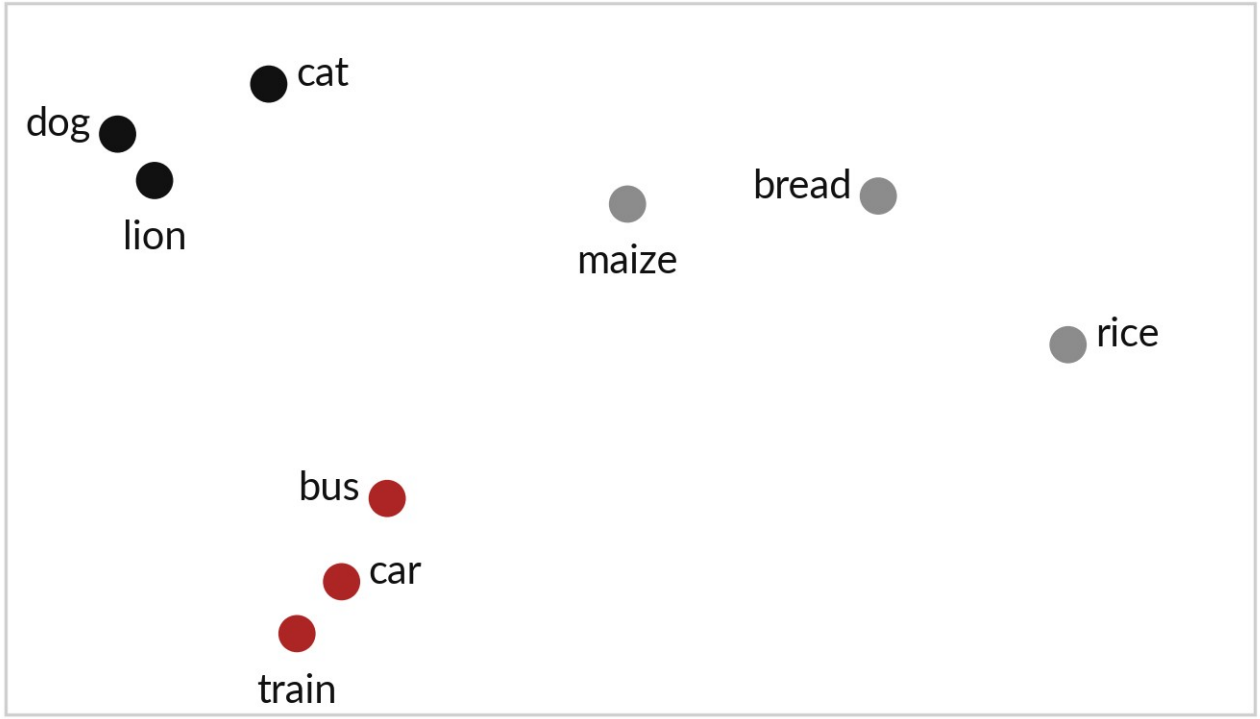
Where this appears in AI: PCA turns embeddings into a map you can look at

Each word is a list of numbers

(illustrative: 4 numbers, made up)

cat	1.00	1.05	0.20	−0.19
car	−0.12	0.15	0.76	0.88
rice	−0.31	−0.87	0.15	−0.93

PCA keeps the two directions with the most spread



principal component 1 (across) and 2 (up)

PCA (principal component analysis) of made-up 4-number vectors. Real embeddings are far longer.

Where this appears in AI: looking at what models learn



PCA keeps the top eigenvectors of the covariance matrix of the data.



The Embedding Projector, published by Google researchers in 2016, offers PCA to view embeddings.



Open-weight models such as Llama publish their weights, so this analysis can be run on them.



Closed assistants do not publish their internal embedding tables. The vectors a public embedding service returns for your own text can be projected the same way.

Where this appears in AI: check an assistant with your numbers

8 minutes

Hands-on, with any AI assistant

Ask an assistant for eigenvalues, then check them.

Ask for the eigenvalues and eigenvectors of the matrix with rows (2, 1) and (1, 2). Expect 3 and 1.

Ask again for the covariance matrix with rows (66, 48) and (48, 38). Expect 102 and 2.

Ask it to show the characteristic equation. Never paste confidential data.

You keep

A note of which answers matched, and one where the assistant used a different sign.

Exit check: three questions with short answers

Question	Short answer
What does $Av = \lambda v$ say in words?	A only stretches, flips or squashes the vector v. λ is the factor.
What are the eigenvalues of the matrix with rows (2, 1) and (1, 2)?	T = 4 and D = 3, so λ = 3 and 1.
What does the top eigenvector of a covariance matrix tell you?	The direction in which the data vary most. Its eigenvalue is the variance there.

Write your own answer first, then compare. Your lab checks are part of the exit check.

Save the lab and keep the four eigenvalues

What you keep from today

Keep the saved lab and four numbers: eigenvalues 3 and 1 for the small matrix, and 102 and 2 for the covariance matrix of the eight marks.

Our question today

In a table with many columns, which directions do the data vary along most, and how do we look at embeddings with hundreds of numbers?

Next session

Session 6: singular value decomposition and PCA, which build on today.

Take away

Saved lab
Notebook or workbook, saved

Small matrix
Eigenvalues 3 and 1

Covariance matrix
Eigenvalues 102 and 2

Next session
SVD and PCA